

# Errata

## Lectures on Electromagnetic Theory

Corrections to the published text. Each entry gives the printed page, the location, the text *as printed*, and the corrected text.

### Chapter 1 — Uniqueness theorem

**p. 21** energy identity and its surface-integral form (and the sentence introducing them).

*Reads:* the difference fields are dotted "... and the second with  $\delta \vec{E}^*$ ," giving

$$\vec{\nabla} \times \delta \vec{E} \cdot \delta \vec{H}^* - \vec{\nabla} \times \delta \vec{H} \cdot \delta \vec{E}^* = \vec{\nabla} \cdot (\delta \vec{E} \times \delta \vec{H}^*) = i\omega (\mu |\delta \vec{H}|^2 - \epsilon |\delta \vec{E}|^2),$$

$$\oint_S (\delta \vec{E} \times \delta \vec{H}^*) \cdot d\vec{s} = i\omega \int_V (\mu |\delta \vec{H}|^2 - \epsilon |\delta \vec{E}|^2) dv.$$

*Should read:* "... and the *conjugate of the second* with  $\delta \vec{E}$ ," giving

$$\vec{\nabla} \times \delta \vec{E} \cdot \delta \vec{H}^* - \vec{\nabla} \times \delta \vec{H}^* \cdot \delta \vec{E} = \vec{\nabla} \cdot (\delta \vec{E} \times \delta \vec{H}^*) = -i\omega (\mu |\delta \vec{H}|^2 - \epsilon^* |\delta \vec{E}|^2),$$

$$\oint_S (\delta \vec{E} \times \delta \vec{H}^*) \cdot d\vec{s} = -i\omega \int_V (\mu |\delta \vec{H}|^2 - \epsilon^* |\delta \vec{E}|^2) dv.$$

### Chapter 2 — Plane waves and waveguides

**p. 42** phase velocity in a low-loss medium.

*Reads:*

$$v_p = \frac{\omega}{\beta} = \left\{ \frac{\mu\epsilon'}{2} \left[ \sqrt{1 + (\tan \delta)^2} - 1 \right] \right\}^{-1/2}.$$

*Should read:*

$$v_p = \frac{\omega}{\beta} = \left\{ \frac{\mu\epsilon'}{2} \left[ \sqrt{1 + (\tan \delta)^2} + 1 \right] \right\}^{-1/2}.$$

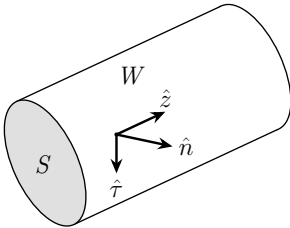
**p. 53** TE-mode boundary condition at a PEC wall.

*Reads:*  $E_\tau^0 = i\omega\mu \partial_n H_z^0$ . *Should read:*  $E_\tau^0 = \frac{i\omega\mu}{k_c^2} \partial_n H_z^0$ .

**p. 54** Figure 2.5 (illustration of a waveguide cross section): direction of the  $\hat{\tau}$  arrow.

*Reads:*  $\hat{\tau}$  is drawn pointing up, which makes the drawn triad  $(\hat{z}, \hat{n}, \hat{\tau})$  left-handed.

*Should read:*  $\hat{\tau}$  should point the opposite way (*downward* along the wall), so that  $\hat{\tau} = \hat{z} \times \hat{n}$  (equivalently  $\hat{n} \times \hat{\tau} = \hat{z}$ ):



**p. 59** modal-orthogonality identity, Cases 1 and 2.

*Reads:*

$$\left(\hat{z} \times \vec{\nabla}_{\perp} \psi_{pq}\right) \cdot \left(\hat{z} \times \vec{\nabla}_{\perp} \psi_{p'q'}\right) = \hat{z} \cdot \left[\hat{z} \left(\vec{\nabla}_{\perp} \psi_{pq} \cdot \vec{\nabla}_{\perp} \psi_{pq}\right)\right] = \vec{\nabla}_{\perp} \psi_{pq} \cdot \vec{\nabla}_{\perp} \psi_{pq}.$$

*Should read:*

$$\left(\hat{z} \times \vec{\nabla}_{\perp} \psi_{pq}\right) \cdot \left(\hat{z} \times \vec{\nabla}_{\perp} \psi_{p'q'}\right) = \hat{z} \cdot \left[\hat{z} \left(\vec{\nabla}_{\perp} \psi_{pq} \cdot \vec{\nabla}_{\perp} \psi_{p'q'}\right)\right] = \vec{\nabla}_{\perp} \psi_{pq} \cdot \vec{\nabla}_{\perp} \psi_{p'q'}.$$

**p. 60** triple-product identity, Case 3.

*Reads:*  $(\hat{z} \times \vec{\nabla}_{\perp} \psi_{pq}) \cdot \hat{n} = \vec{\nabla}_{\perp} \psi_{pq} \cdot (\hat{z} \times \hat{n}) = -\vec{\nabla}_{\perp} \psi_{pq} \cdot \hat{\tau}.$

*Should read:*  $(\hat{z} \times \vec{\nabla}_{\perp} \psi_{pq}) \cdot \hat{n} = \vec{\nabla}_{\perp} \psi_{pq} \cdot (\hat{n} \times \hat{z}) = -\vec{\nabla}_{\perp} \psi_{pq} \cdot \hat{\tau}.$

## Chapter 3 — Radiation

**p. 74** far-field approximation, order of the binomial expansion.

*Reads:* "... Substituting into  $(1+x)^{1/2} \approx 1 + \frac{1}{2}x$  and multiplying by  $r$  gives  $R \approx r - \hat{r} \cdot \vec{r}' + \frac{r'^2 - (\hat{r} \cdot \vec{r}')^2}{2r} \dots$ "

*Should read:* "... Substituting into  $(1+x)^{1/2} \approx 1 + \frac{1}{2}x - \frac{1}{8}x^2 \dots$ " (the displayed second-order  $R$  requires the  $-\frac{1}{8}x^2$  term; with  $1 + \frac{1}{2}x$  alone the result would be  $R \approx r - \hat{r} \cdot \vec{r}' + r'^2/2r$ ).

**p. 76** Hertzian dipole, the intermediate  $\vec{\nabla}[\vec{\nabla} \cdot (\hat{z}G)]$ .

*Reads:*

$$\vec{\nabla}[\vec{\nabla} \cdot (\hat{z}G)] = G \left[ \left( -k^2 + \frac{2ik}{r} + \frac{1}{r^2} \right) \cos \theta \hat{r} + \left( ik + \frac{1}{r} \right) \sin \theta \hat{\theta} \right].$$

*Should read:*

$$\vec{\nabla}[\vec{\nabla} \cdot (\hat{z}G)] = G \left[ \left( -k^2 + \frac{2ik}{r} + \frac{2}{r^2} \right) \cos \theta \hat{r} + \frac{1}{r} \left( ik + \frac{1}{r} \right) \sin \theta \hat{\theta} \right].$$

**p. 82** rectangular-aperture far field, phase factor.

*Reads:*  $\dots e^{ik(x'u+y'v)} dx' dy'.$  *Should read:*  $\dots e^{ik(x'u+y'v)} dx' dy'.$

**p. 82** rectangular-aperture far field, amplitude prefactor.

*Reads:*  $\vec{E} \approx ikE_0ab \frac{e^{-ikr}}{4\pi r} (\dots) \text{sinc}\left(\frac{ka}{2}u\right) \text{sinc}\left(\frac{kb}{2}v\right).$

*Should read:*  $\vec{E} \approx ikE_0ab \frac{e^{-ikr}}{2\pi r} (\dots) \text{sinc}\left(\frac{ka}{2}u\right) \text{sinc}\left(\frac{kb}{2}v\right).$

## Chapter 4 — Electromagnetic scattering

**p. 90** cylindrical-wave (Jacobi–Anger) expansion, projection step.

*Reads:* "Multiplying [the expansion] by  $e^{im\phi}$  and integrating over  $\phi \dots$ "

*Should read:* "Multiplying [the expansion] by  $e^{-im\phi}$  and integrating over  $\phi \dots$ " (the projector must be  $e^{-im\phi}$  to match the two stated integral identities; the result  $a_m = i^{-m}$  is unchanged).

**p. 96** separation of variables, scalar Helmholtz equation.

*Reads:* "... multiply  $r^2 \sin \theta$  and divide it by  $\psi$  ..." *Should read:* "... multiply  $r^2 \sin^2 \theta$  and divide it by  $\psi$  ..."

**p. 97** Legendre regularity condition.

*Reads:* "... remain finite at  $\sin \theta = \pm 1$  ..." *Should read:* "... remain finite at  $\cos \theta = \pm 1$  ..."

**p. 100** incident-field Debye-potential expansion.

*Reads:*  $\pi_e^i = \sum_{n=0}^{\infty} \sum_{m=0}^n j_n(k_2 r) P_n^m(\cos \theta) (A_{mn} \cos m\phi + B_{mn} \sin m\phi)$ .

*Should read:*  $\pi_e^i = \sum_{n=0}^{\infty} \sum_{m=0}^n j_n(kr) P_n^m(\cos \theta) (A_{mn} \cos m\phi + B_{mn} \sin m\phi)$ .

**p. 101** Riccati-Bessel identity.

*Reads:*  $\frac{d^2}{dr^2} [r j_n(kr)] + (kr)^2 j_n(kr) = n(n+1) j_n(kr)$ .

*Should read:*  $\left( \frac{d^2}{dr^2} + k^2 \right) [r j_n(kr)] = \frac{n(n+1)}{r} j_n(kr)$ .

**p. 98 and 104** second-kind spherical Hankel function (two occurrences).

*Reads:*

$$h_n^{(2)}(kr) = \sqrt{\frac{\pi}{2kr}} H_{n+1/2}(kr), \quad \hat{h}_n^{(2)}(kr) = \sqrt{\frac{\pi kr}{2}} H_{n+1/2}(kr) \approx i^{n+1} e^{-ikr}.$$

*Should read:* the cylindrical Hankel must carry the second-kind superscript  $H_{n+1/2}^{(2)}$ :

$$h_n^{(2)}(kr) = \sqrt{\frac{\pi}{2kr}} H_{n+1/2}^{(2)}(kr), \quad \hat{h}_n^{(2)}(kr) = \sqrt{\frac{\pi kr}{2}} H_{n+1/2}^{(2)}(kr).$$

**p. 104** dielectric-sphere matching conditions ( $\epsilon$  and  $\mu$  interchanged in the algebraic continuity of the Debye potentials).

*Reads:*

$$\mu \pi_e = \mu_d \pi_e^d, \quad \epsilon \pi_{mr} = \epsilon_d \pi_m^d.$$

*Should read:*

$$\epsilon \pi_e = \epsilon_d \pi_e^d, \quad \mu \pi_{mr} = \mu_d \pi_m^d.$$

Continuity of the tangential  $\vec{H}$  carries the factor  $i\omega\epsilon\Pi_e$  and tangential  $\vec{E}$  the factor  $i\omega\mu\Pi_m$ ; the corrected assignment is the one that reproduces the printed coefficients  $a_n, b_n$ .

**p. 105** the  $|S_1(\theta)|^2$  and  $|S_2(\theta)|^2$  expansions (bodies interchanged, and one mis-paired term).

With  $c_n = (2n+1)/[n(n+1)]$ :

*Reads:*

$$|S_1|^2 = \sum_{n,n'} c_n c_{n'} [a_n \tau_n a_{n'}^* \tau_{n'} + b_n \pi_n b_{n'}^* \pi_{n'} + a_n \tau_n b_{n'}^* \pi_{n'} + b_n \pi_n a_{n'}^* \tau_{n'}],$$

$$|S_2|^2 = \sum_{n,n'} c_n c_{n'} [a_n \pi_n a_{n'}^* \pi_{n'} + b_n \tau_n b_{n'}^* \tau_{n'} + a_n \pi_n b_{n'}^* \tau_{n'} + b_n \tau_n a_{n'}^* \pi_{n'}].$$

Should read:

$$|S_1|^2 = \sum_{n,n'} c_n c_{n'} [a_n \pi_n a_{n'}^* \pi_{n'} + b_n \tau_n b_{n'}^* \tau_{n'} + a_n \pi_n b_{n'}^* \tau_{n'} + b_n \tau_n a_{n'}^* \pi_{n'}],$$

$$|S_2|^2 = \sum_{n,n'} c_n c_{n'} [a_n \tau_n a_{n'}^* \tau_{n'} + b_n \pi_n b_{n'}^* \pi_{n'} + a_n \tau_n b_{n'}^* \pi_{n'} + b_n \pi_n a_{n'}^* \tau_{n'}].$$

**p. 105**  $\pi/\tau$  orthogonality normalization.

Reads:  $\int_0^\pi (\pi_n \pi_{n'} + \tau_n \tau_{n'}) \sin \theta d\theta = \frac{2n(n+1)^2}{2n+1} \quad (n = n').$

Should read:  $\int_0^\pi (\pi_n \pi_{n'} + \tau_n \tau_{n'}) \sin \theta d\theta = \frac{2n^2(n+1)^2}{2n+1} \quad (n = n').$

**p. 106** PEC-sphere Mie coefficients.

Reads:

$$a_n = -\frac{\hat{j}'_n(ka)}{\hat{h}_n^{(2)'}(ka)}, \quad b_n = -\frac{\hat{j}_n(ka)}{\hat{h}_n^{(2)}(ka)}.$$

Should read:

$$a_n = +\frac{\hat{j}'_n(ka)}{\hat{h}_n^{(2)'}(ka)}, \quad b_n = +\frac{\hat{j}_n(ka)}{\hat{h}_n^{(2)}(ka)}.$$

**p. 106** dielectric-sphere internal Mie coefficients  $c_n, d_n$  (numerators).

Reads:

$$c_n = \frac{i\sqrt{\epsilon_d \mu}}{D_e}, \quad d_n = \frac{i\sqrt{\epsilon \mu_d}}{D_m},$$

Should read:

$$c_n = \frac{i\mu_d \sqrt{\epsilon_d/\mu}}{D_e}, \quad d_n = \frac{i\mu_d \sqrt{\epsilon_d/\mu}}{D_m},$$

where  $D_e, D_m$  are the (unchanged) denominators that appear in  $a_n, b_n$ .

**p. 107** Rayleigh (small-sphere) scattering coefficients, overall sign.

Reads:

$$a_1 = -\frac{2i}{3}(ka)^3 \frac{\epsilon_d - \epsilon}{\epsilon_d + 2\epsilon}, \quad b_1 = -\frac{2i}{3}(ka)^3 \frac{\mu_d - \mu}{\mu_d + 2\mu}.$$

Should read:

$$a_1 = +\frac{2i}{3}(ka)^3 \frac{\epsilon_d - \epsilon}{\epsilon_d + 2\epsilon}, \quad b_1 = +\frac{2i}{3}(ka)^3 \frac{\mu_d - \mu}{\mu_d + 2\mu}.$$

## Chapter 5 — Special relativity

**p. 111** derivation of the Lorentz transformation.

Reads: “Since by **(P2)** it is required that the transformation is linear ...” Should read: “Since by **(P1)** it is required that the transformation is linear ...”

## Appendix B — Bessel and Hankel functions

**p. 133** physical roles of the Hankel functions.

Reads: “... representing *outward and inward* cylindrical waves, respectively.”

Should read: “... representing *inward and outward* cylindrical waves, respectively.”

## Appendix C — Legendre functions

**p. 139** Legendre polynomial  $P_5(x)$ , missing factor of  $x$ .

*Reads:*  $P_5(x) = \frac{1}{8}(63x^5 - 70x^3 + 15)$ . *Should read:*  $P_5(x) = \frac{1}{8}(63x^5 - 70x^3 + 15x)$ . (The last term must carry the factor  $x$ ; as printed,  $P_5$  is not an odd function and gives  $P_5(0) = \frac{15}{8} \neq 0$ , contradicting properties (i) and (iii).)

**p. 140** Legendre function of the second kind, closed form.

*Reads:*

$$Q_n(x) = \frac{1}{2}P_n(x) \ln\left(\frac{1+x}{1-x}\right) - \sum_{k=1}^n \frac{1 \cdot 3 \cdot 5 \cdots (2k-1)}{2 \cdot 4 \cdot 6 \cdots (2k)} P_{n-k}(x).$$

*Should read:*

$$Q_n(x) = \frac{1}{2}P_n(x) \ln\left(\frac{1+x}{1-x}\right) - \sum_{k=1}^n \frac{1}{k} P_{k-1}(x) P_{n-k}(x).$$